XUAN-NGHIA HUYNH

PHD CANDIDATE

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AI PhD candidate specializing in computer vision (object detection, segmentation) and generative AI (GANs, diffusion models), focused on layout-conditioned and multi-spectral image synthesis with automated annotation. Seeking full-time roles in AI Research, Computer Vision, or Generative AI from December 2026.

EDUCATION

PhD Candidate in AI & Robotics — *Sejong University* 03/2022 – Present, Seoul, South Korea

Dissertation (ongoing): Generative Frameworks for Layout-Conditioned RGB and Multi-Spectral Fire Image Synthesis to Enhance Detection and Segmentation Tasks.

GPA 4.41 / 4.5 **Expected Graduation: Feb 2027**

B.E. Electrical & Electronics Engineering — *HCMUT, VNU-HCM* 09/2017 – 08/2021, Ho Chi Minh City, Vietnam

Thesis: Mobile Robot Applied 3D SLAM & Deep Learning.
Honor Class of Control Engineering & Automation.

GPA 8.56 / 10

PROFESSIONAL EXPERIENCE

PhD Researcher — 03/2022 – Present

IVPG Laboratory, Sejong University · Seoul, South Korea

Small Object Detection for Real-Time Embedded Systems (First Author)

- Proposed **YOLOv4-SRm**, a super-resolution module improving AP50 by **+11.38%** (company data) and **+6.29%** (VisDrone2019) vs. vanilla YOLOv4, with fewer parameters.
- Pruned & quantized a simplified variant (SSRm) and deployed it on a Qualcomm QCS610 SoC at **36 FPS** for a real-time CCTV detection framework.

GAN-based Layout-Guided Fire Image & Mask Generation (First Author)

- Built a layout-conditioned GAN (Image-to-Image + Layout-to-Image) generating paired fire images and segmentation masks with no manual annotation; improved FID by **4.79** vs. PaintByExample.
- Validated via data augmentation: **+4.6% mAP** (YOLOv9 detection) and **+4.44% mIoU** (U-Net segmentation).

Diffusion-based Multi-Spectral Fire Synthesis (First Author, ongoing)

- Developed a layout-conditioned diffusion model with a layout-aware cross-attention mechanism for RGB-only and RGB-NIR fire synthesis.
- Proposed a pixel-cross attention module for automatic mask refinement during synchronized RGB+NIR generation.

Undergraduate Research Assistant — 01/2020 – 08/2021

Autonomic Control Lab, HCMUT · Ho Chi Minh City, Vietnam

- Designed and built a three-wheel mobile robot (SolidWorks, ROS1) running RTAB-Map 3D SLAM with AMCL localization, Dijkstra path planning, and Dynamic Window obstacle avoidance.
- Integrated 3D SLAM with a deep-learning human-tracking model via a Wi-Fi client–server architecture.

Embedded Software Engineer (Intern) — 06/2020 – 12/2020

Robert Bosch Engineering and Business Solutions VN · Ho Chi Minh City, Vietnam

- RADAR team: configured AUTOSAR RTE interfaces/ports and implemented CAN RX message fault monitoring (DLC, timeout, checksum, alive-counter checks) against a CAN Matrix/DBC spec.

SKILLS

Generative AI
Diffusion Models (DDPM, DDIM, Latent Diffusion), GANs (CycleGAN, SPADEGAN, LostGAN), cross-spectral translation, transformer attention, layout-conditioning

Computer Vision
Object detection, semantic segmentation, YOLO (v4, v9), knowledge distillation, multi-GPU training, CNNs, MLOps

Research
Analyzing and implementing English-language papers, writing papers for international journals & conferences, research and algorithm development

Programming & Frameworks
Python (PyTorch, TorchVision, OpenCV, Pandas, NumPy, TensorFlow), C/C++

Tools
Git, Docker, Linux, Jupyter, Hugging Face, Microsoft Azure, Isaac Sim

Languages
Vietnamese (Native), English (Fluent, IELTS 7.0), Korean (Basic)

PUBLICATIONS

Fire Image Generation Based on Image-to-Image and Layout-to-Image Translations for Detection and Segmentation Tasks
IEEE Access · 2026

Diffusion-based Fire Image Synthesis for Generating Object Detection Datasets
Proc. 2025 Summer Conf. of IEIE · 2025

One-Stage Small Object Detection Using Super-Resolved Feature Map for Edge Devices
MDPI Electronics · 2024

REFERENCE

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